

Program : Diploma in Tool & Die Engineering	
Course Code : 5117	Course Title: Computer Aided Press Tool Drawing
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart knowledge on detailed part drawings and assembly drawings of various types of press tools from the given component drawing with the aid of design principles learned from press tool technology.
- To impart skills in drafting and in the creation of 3D models using Auto CAD/PTC Creo/solidworks/NX Unigraphics/CATIA/Delcam software etc.

Course Prerequisites:

Topic	Course code	Course name	Semester
Design concepts of press tool		Press Tool Technology	4
Limits, Fits, G&D Tolerance, Surface Finish symbols and values		Tool Engineering Drawing	3
Heat treatment processes & various tool steels		Material Science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare detailed drawings of guide plate & blanking tools.	9	Applying
CO2	Prepare detailed drawings of Progressive Tools and Advanced Multi stage Tools.	12	Applying

CO3	Prepare detailed drawings of compound & Combination Tools	12	Applying
CO4	Prepare detailed drawings of Bending, Forming & Drawing dies	10	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			
CO2	3		3	3			3
CO3	3		3	3			
CO4	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Prepare detailed drawings of guide plate and blanking tools.		
M1.01	Design single stage guide plate tool	3	Applying
M1.02	Design three stage guide plate tool		Applying
M1.03	Design 3 stage progressive tool with die set	6	Applying
	3D Modelling and drafting shall be done by using any of the above-mentioned software.		
CO2	Prepare detailed drawings of Progressive Tools and Advanced Multi stage Tools.		
M2.01	Design a progressive tool with travelling stripper.	3	Applying
M2.02	Design a progressive tool with side cutter and travelling stripper.	3	Applying
M2.03	Design a progressive tool incorporating parting off operation.	3	Applying
M2.04	Design a Progressive Tool with multiple operation- cutting, bending, and side cam piercing and parting.	3	Applying
	3D Modelling and drafting shall be done by using any of the above-mentioned software.		
	Series Test – I	1	

CO3	Prepare detailed drawings of compound & Combination Tools		
M3.01	Design a compound tool with direct knock out mechanism	3	Applying
M3.02	Design a compound tool with indirect knock out mechanism	3	Applying
M3.03	Design a draw and pinch trim combination tool	6	Applying
	3 D Modelling and drafting shall be done by using any of the above-mentioned software.		
CO4	Prepare detailed drawings of Bending, Forming & Drawing dies		
M 4.01	Design a 'V' bending tool	3	Applying
M 4.02	Design a 'L' bending tool		Applying
M 4.03	Design a 'U' bending tool	3	Applying
M 4.04	Design a 'Z' bending tool		Applying
M 4.05	Design a simple forming/coining/embossing die	4	Applying
M 4.06	Design a draw tool with mechanical blank holding.		Applying
	3 D Modelling and drafting shall be done by using any of the above-mentioned software.		
	Series Test – II	1	

Text/ Reference:

T/R	Book Title/Author
T1	J.R.Paquin, Die design fundamentals, Industrial Press Inc, 1990.
R2	Donaldson, Tool Design, Tata McGraw-hill Book company, 23rd edition, 2006
R3	Indian Standard Specifications IS 3455, IS 6137, IS 6244, IS 6246, IS 7018: Part 2, IS 3477 and IS 3485

Online Resources:

Sl.No	Website Link
1	https://www.ijert.org/research/design-and-analysis-of-progressive-tool-IJERTV1IS6067.pdf
2	https://www.ijntr.org/download_data/IJNTR03040039.pdf
3	https://www.misumi-techcentral.com/tt/en/press/2013/04/155-design-of-compound-punching-dies-4-structure-of-a-compound-punching-die.html
4	http://www.ijrerd.com/papers/v2-i1/6.IJRERD-B015.pdf
5	https://www.ijsr.net/archive/v3i5/MjMwNDE0MDE=.pdf
6	https://www.machinemfg.com/metal-stamping-and-die-design-bending/